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Mapping Mangrove Forest from LiDAR Data Using Object-Based Image Analysis and Support Vector Machine: The Case of Calatagan, Batangas

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***Abstract*—Efforts have been made in conserving the mangrove forests of Calatagan, Batangas. As an aid in the conservation and environmental assessment of the said forest, this study presents a method in producing fine scale map of these protected areas. The study site covers the northern part of Calatagan consisting of 12 barangays, seven of which are in the coastal zone. Serving as input in image classification, airborne LiDAR (Light Detection and Ranging) data measuring approximately 72 km^2 was provided by DREAM LiDAR. Combinations of multi-threshold and multiresolution segmentation enabled the pre-processing stage before moving on with the classification. It was observed that non-ground features are best segmented by giving weights to pit-free CHM, average intensity and number of returns arithmetic. Rule-based approach was implemented in identifying artificial surfaces such as buildings and fences; then, the Support Vector Machine (SVM) algorithm was applied in classifying the remaining non-ground objects. Fourteen features were selected for the training phase of SVM: eight from LiDAR-derived layers (number of returns, vertical features and intensity) and six from geometry features. Applying accuracy assessment based on test and training area (TTA) mask derived from validation points, the producer's and user's accuracy for mangrove, artificial surfaces and other vegetation were above 0.87. As a result, the overall accuracy of the produced fine scale mangrove map was 0.91 with a Kappa Index of Agreement of 0.85. The mapping product can then be used by local government unit and management agencies as a reference in baseline assessment of mangrove resources, and in planning and implementing further conservation efforts.**

Keywords—LiDAR; mangrove mapping; support vector machine; object-based image analysis

I. Introduction

Mangrove forests are commonly found on flats along the sea coast, extending along the stream where the water is brackish. It has been recognized that mangroves have significant ecological functions in protecting the coastlines of sheltered tropical regions. Mangroves can attenuate waves, storm surges or tsunamis [1]. As found by Zhang, et al. [2] in their field observations and numerical simulations, mangroves of 6-to-30-km-wide reduced the impact of storm surges from a Category 3 hurricane along the Gulf Coast of South Florida, thereby protecting the inland wetland by decreasing the inundation area. Another important role of mangroves is in providing nutrients for its neighboring estuarine and coastal ecosystems. Kon, et al. [3] have shown that the physical structure of mangrove forest facilitates habitation of intertidal benthic faunal communities.

In 1950, the mangrove area of Calatagan covered 132.8 hectares; however, a study in 2001 indicated that the area thinned to 18 hectares [4]. Recognizing its important role, efforts have been made in conserving the mangrove forests of Calatagan, Batangas. In fact, the local government of Calatagan inked Sangguniang Bayan Resolution No. 76 on 8 September 2009 creating the Calatagan Mangrove Forest Conservation Park centering in Brgy. Quilitisan. As an aid in the conservation and environmental assessment of the said mangrove forest, this study presents a method in producing fine scale map of these protected areas.

Traditionally, broad scale map of mangroves can be derived from the available of Landsat image, with resolution ranging between 15-30 m. With the advent of LiDAR technology in the Philippines, a 1-m resolution thematic map of mangrove can be generated. This study is aligned with the objective of Phil-LiDAR2 Project 2: Aquatic Resources Extraction from LiDAR Surveys, which is to map mangrove, aquaculture and benthic habitats.

A problem that arises from data processing is the availability of orthophotograph. LiDAR data combined with orthophoto can improve the accuracy of classification [5]. However, most areas surveyed do not have such aerial image. Thus, this study is aimed at developing methods to optimize the LiDAR data available to attain the project's objective.

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II. Methodology

A. Study Area

Calatagan town in Batangas, Philippines (Fig. 1) has a 48-km coastline; 19 of its 25 barangays are in coastal areas. The study site covers 12 barangays, 7 of which are in the coastal zone. Besides mangroves, other dominant tree species are mango and coconut. Agricultural produce also include sugarcane and rice. Aquaculture of fish ponds and prawn ponds are also common.

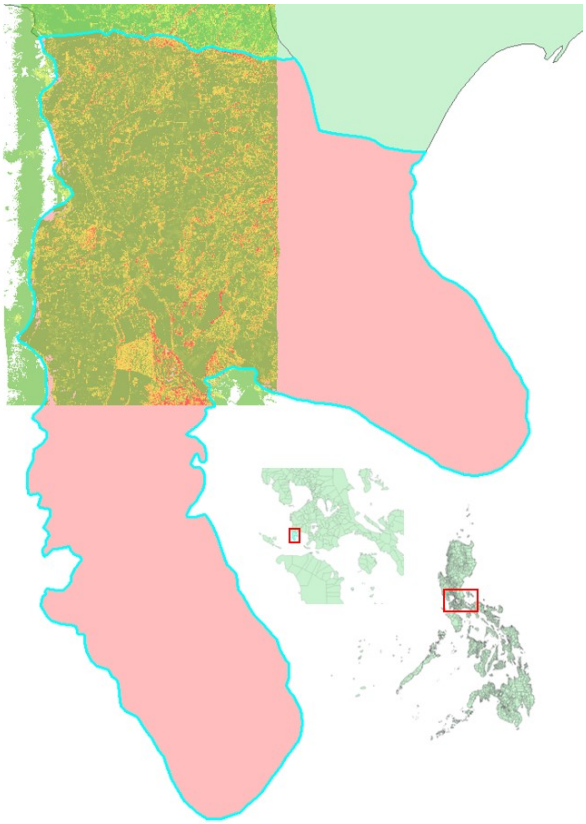

Fig. 1. The study area in Calatagan, Batangas with canopy height model superimposed on the municipality's shapefile.

B. Materials

Sourced from the DREAM-LiDAR (Disaster Risk and Exposure Assessment for Mitigation-Light Detection and Ranging), topographic airborne LiDAR served as input in image classification, measuring approximately 72 km^2. The area was surveyed on 20 February 2014. Fig. 2 (b-d) shows some screenshot of the dataset used in the classification.

We generated various LiDAR derivatives (Fig. 3) using LAStools. The image segmentation and classification were done in eCognition 9.0.

C. Classes

Fig. 4 shows the general workflow for extracting mangroves from the raw data. We classified dataset into 6 classes:

- Water
- Sea
- Ground
- Mangrove
- Non-vegetation (includes building and fences)
- Other vegetation (includes sugarcane, mango, etc.)

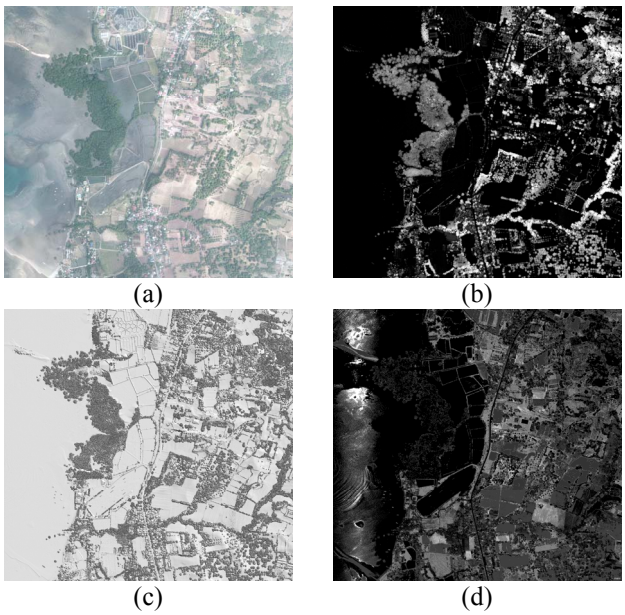

Fig. 2. (a) Sample orthophotograph of the study site, (b) pit-free canopy height model, (c) digital surface model with hillside shading and (d) average intensity.

D. Support Vector Machine

A non-parametric supervised classification, Support Vector Machine (SVM) classifier is derived from statistical learning theory; its foundation was developed by Vapnik (1995). SVM does not assume that the data is normally distributed in the image. Classification task involves separating data into training and testing. SVM aims to produce a model (based on the training data) which predicts the target values of the test data given only the test data attributes [6].

SVM finds many applications, especially in remote sensing. Using time-series Landsat data, Zheng et al. [7] successfully classified nine irrigated crop types using SVM, whose overall accuracy were recorded at >86%. It is important to note that intelligent selection approach can reduce number of training samples and still achieve higher overall accuracy. This capability of the SVM can enable classification even if training points from the field is scarce.

Lodha et al. [8] also used SVM algorithm but this time in classifying objects from airborne LiDAR data. They used five LiDAR derivatives which includes height, height variation, normal variation, intensity and number of returns.

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They obtained an overall accuracy of more than 90% with convincing visual results.

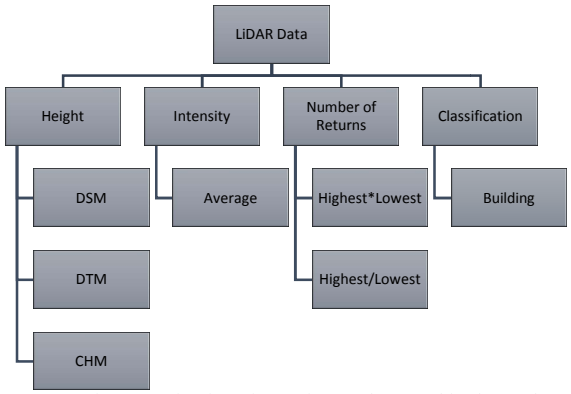

Fig. 3. Derivatives from LiDAR data used in the study.

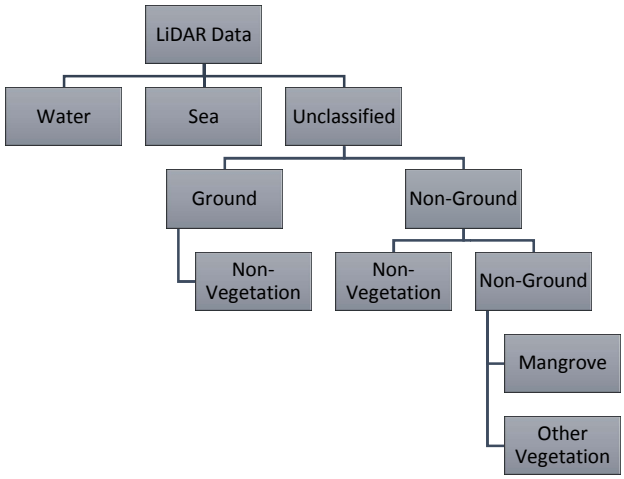

Fig. 4. General workflow of extracting mangroves from the dataset.

E. Features and Training

From the LiDAR data, we derived and identified layers used in the image segmentation and data classification. Based on height information, digital surface model (DSM – actual values and with hillside shading), digital terrain model (DTM) and pit-free canopy height model (CHM) were extracted. Alongside, the average intensity was also taken from the raw data. Intensity records the energy of the response reflected from the terrain to the laser scanner. With classification based on point clouds, a building raster was generated to lessen the confusion during image classification.

The LiDAR data also recorded multiple returns ranging from 1-4. From these returns, we have taken the highest and lowest number of returns and did raster calculation. An important finding noted that LiDAR return ratio features can distinguish tree species because of variations in canopy density [9]. For instance, we found that sugarcane canopy has approximately a value of 2 for both highest*lowest and highest/lowest number of returns arithmetic.

Meanwhile, we gathered training and validation points in the area on 1-3 June 2015. Even if one year has passed since the flight of airborne LiDAR, almost no change were detected when we surveyed the area, particularly in mangroves area.

F. Methods

Fig. 5 shows the developed rule set in eCognition. Rule-based approach was first implemented to identify water and land by using the DSM as input layer in multi-threshold segmentation. Alternatively, we can use the DTM to separate the land and water. Using contextual editing, large body of water (with area greater than 1,000,000 pixels) was assigned as sea.

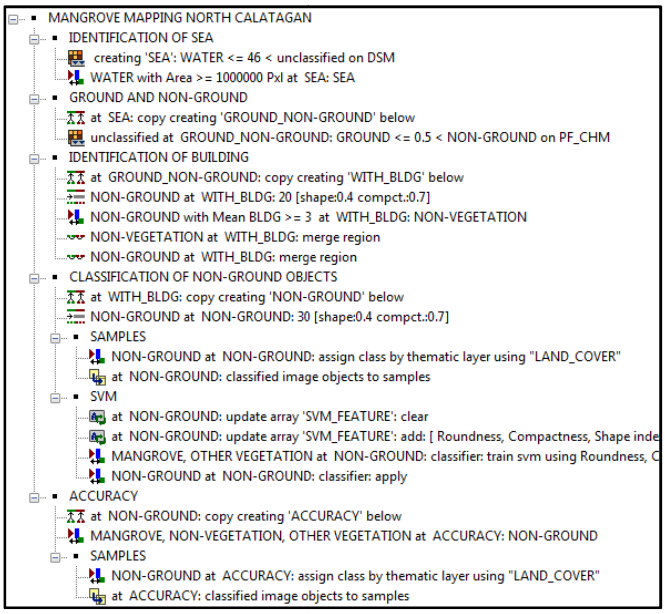

Fig. 5. Rule set developed in the study

To separate ground and non-ground, we used the pit-free CHM as input layer in doing multi-threshold segmentation because it has normalized height with respect to ground. Afterwards, artificial surfaces were classified on the segmented non-ground objects with the aid of the building raster. Then, we let the SVM algorithm to classify the remaining non-ground objects into two: mangrove and other vegetation. It is important to note that we did not classify the other vegetation on the ground objects. Only objects whose height are above 0.5 m based on CHM.

III. Results and Discussion

Compared with contrast-split segmentation, multi-threshold exhibits quicker segmentation and separation of ground and non-ground objects in terms of processing time. Because of errors inherent on the raw data, few misclassification were observed, especially in bodies of water. Nonetheless, it would not affect the classification of mangrove.

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Multiresolution segmentation was applied for non-ground objects before classification. We found that the said objects are best segmented by giving weights to pit-free CHM, average intensity and number of returns arithmetic. The scale used for final classification was 30, shape of 0.4 and compactness equal to 0.7.

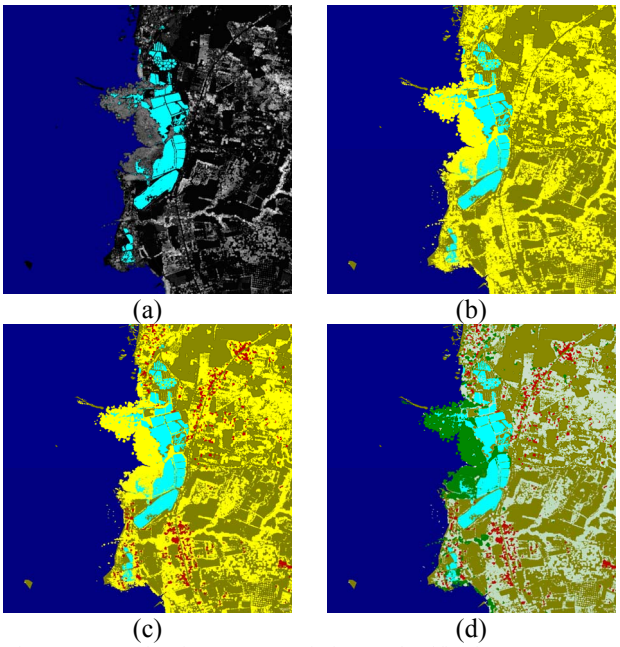

Fig. 6. Output of various processes in image classification: (1) water-cyan, sea-blue and land-gray; (b) water, sea, ground-brown and non-ground-yellow; (c) water, sea, ground, non-ground and artificial-red; and (d) water, sea, artificial, other vegetation-light green and mangrove-green.

TABLE 1. ERROR MATRIX BASED ON TTA MASK

User \ Reference Class	Non-Vegetation	Other Vegetation	Mangrove	Sum
Confusion Matrix				
Non-Vegetation	57168	7864	0	65032
Other Vegetation	8548	219843	16736	245127
Mangrove	0	8311	155591	163902
unclassified	0	0	0	0
Sum	65716	236018	172327	
Accuracy				
Producer	0.8699	0.9315	0.9029	
User	0.8791	0.8969	0.9493	
Hellden	0.8745	0.9138	0.9255	
Short	0.7770	0.8413	0.8613	
KIA Per Class	0.8492	0.8581	0.8516	
Totals				
Overall Accuracy	0.9125			
KIA	0.8537			

Fourteen features were selected for the training phase of SVM: eight from LiDAR-derived layers (DSM, DSM will hillside shading, DTM, average intensity, highest number of returns*lowest number of returns, highest number of returns/lowest number of returns, slope and pit-free CHM) and six from geometry features (roundness, shape index, rectangular fit, compactness, density and elliptic fit). As input parameters, cost parameter used was 200, while kernel spread function was 0.

After classifying objects from the training points, we exported the object statistics for mangroves. We computed the mean, standard deviation and coefficient of variations of all the input features.

For the layers, it is important to note that DTM, DSM, average intensity and slope had coefficient of variations less than 13%. DTM has the lowest CV of 0.65%. Because mangroves are generally found on flat occurring along the coast, the values of DTM and DSM are lower compared with other vegetation in the hilly terrain of Calatagan. The mean ratio of highest number of returns and lowest number of returns for mangroves is 1.89, with CV equal to 26%. This feature increased the accuracy of classification.

Applying accuracy assessment based on test and training area (TTA) mask derived from validation points, the producer's and user's accuracy for mangrove, artificial surfaces and other vegetation were above 0.87 [Table 1]. As a result, the overall accuracy of the produced fine scale mangrove map was 0.91 with a Kappa Index of Agreement of 0.85.

Some confusion between mangroves and other vegetation might be attributed to the inclusion of the pit-free CHM in the image classification because it has the highest value of coefficient of variation (46.57%). Because we considered both matured and young mangroves in getting the training points, the variation in its height is greater; thus, it can really be confused with other vegetation having similar height feature. Among the geometry features, roundness has the highest CV equal to 27.61%, while all others have CV less than 11%.

IV. CONCLUSION AND FUTURE WORK

Even with the absence of spectral information, a fine scale map of mangrove can still be generated by using various LiDAR derivatives. Object-based image analysis proves to be useful in segmenting meaningful objects from the LiDAR layers with certain homogeneity that can distinguish one class from another class using multiresolution segmentation algorithm. The object-based approach can be repeatable. Although the approach here has mapped majority of the mangrove area, it did not entirely map small areas whose mangrove heights were below 0.5 m based on the normalized digital surface model.

With a lot of input features available, the SVM classifier can properly classify mangroves and other features found in the LiDAR data. We recommend optimization of the cost parameter and the kernel spread function in able to achieve higher accuracy.

Importantly, the user's accuracy and producer's accuracy for the specie of concern are above 90%, which satisfies the objective of mapping mangroves accurately. Using a three-scheme classification, the overall accuracy of the map recorded 0.9125 and KIA of 0.8537.

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The mapping product can then be used by local government unit and management agencies as a reference in baseline assessment of mangrove resources, and in planning and implementing further conservation efforts.

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REFERENCES

[1] E. Barbier, E. Koch, B. Silliman, S. Hacker, E. Wolanski, J. Primavera, E. Granek, S. Polasky, S. Aswani and L. Cramer, "Coastal ecosystem-based management with nonlinear ecological functions and values. , 321.," *Science 319,* p. 321, 2008.

[2] K. Zhang, H. Liu, Y. Li, H. Xu, J. Shen and J. Rhome, "The role of mangroves in attenuating storm surges," *Estuarine, Coastal and Shelf Science 102-103,* pp. 11-23, 2012.

[3] K. Kon, H. Kurokura and P. Tongnunui, "Effects of the physical structure of mangrove vegetation on a benthic faunal community," *Journal of Experimental Marine Biology and Ecology 383,* pp. 171-180, 2010.

[4] A. R. Quicoy and N. D. Briones, "Beach Carrying Capacity Assessment of Coastal Ecotourism in Calatagan, Batangas, Philippines," *Journal of Environmental Science and Management 12 (2),* pp. 11-27, 2009.

[5] L. C. G. David, S. Sarte, F. Caluyo and A. J. Ballado, "Fusion of LiDAR Data and Orthophoto to Classify Land Use and Land Cover Using Object-Based Image Analysis," in *Philippine Geomatics Symposium*, Manila, 2014.

[6] C.-W. Hsu, C.-C. Chang and C.-J. Lin, "A Practical Guide to Support Vector Classification," National Taiwan University, Taipei, Taiwan, 2010.

[7] B. Zhenga, S. W. Myinta, P. S. Thenkabailb and R. M. Aggarwal, "A support vector machine to identify irrigated crop types using time-series Landsat NDVI data," *International Journal of Applied Earth Observation and Geoinformation 34,* pp. 103-112, 2015.

[8] S. K. Lodha, E. J. Kreps, D. P. Helmbold and D. Fitzpatrick, "Aerial LiDAR Data Classification using Support Vector Machines," in *Third International Symposium on 3D Data Processing, Visualization, and Transmission*, 2006.

[9] S. Takeshi, I. Junichi, I. Keiko, M. Yukihiro and K. Katsunori, "Object-based classification of land cover and tree species by integrating airborne LiDAR and high spatial resolution imagery data," *Landscape Ecol. Eng.,* pp. 157-171, 2012.

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